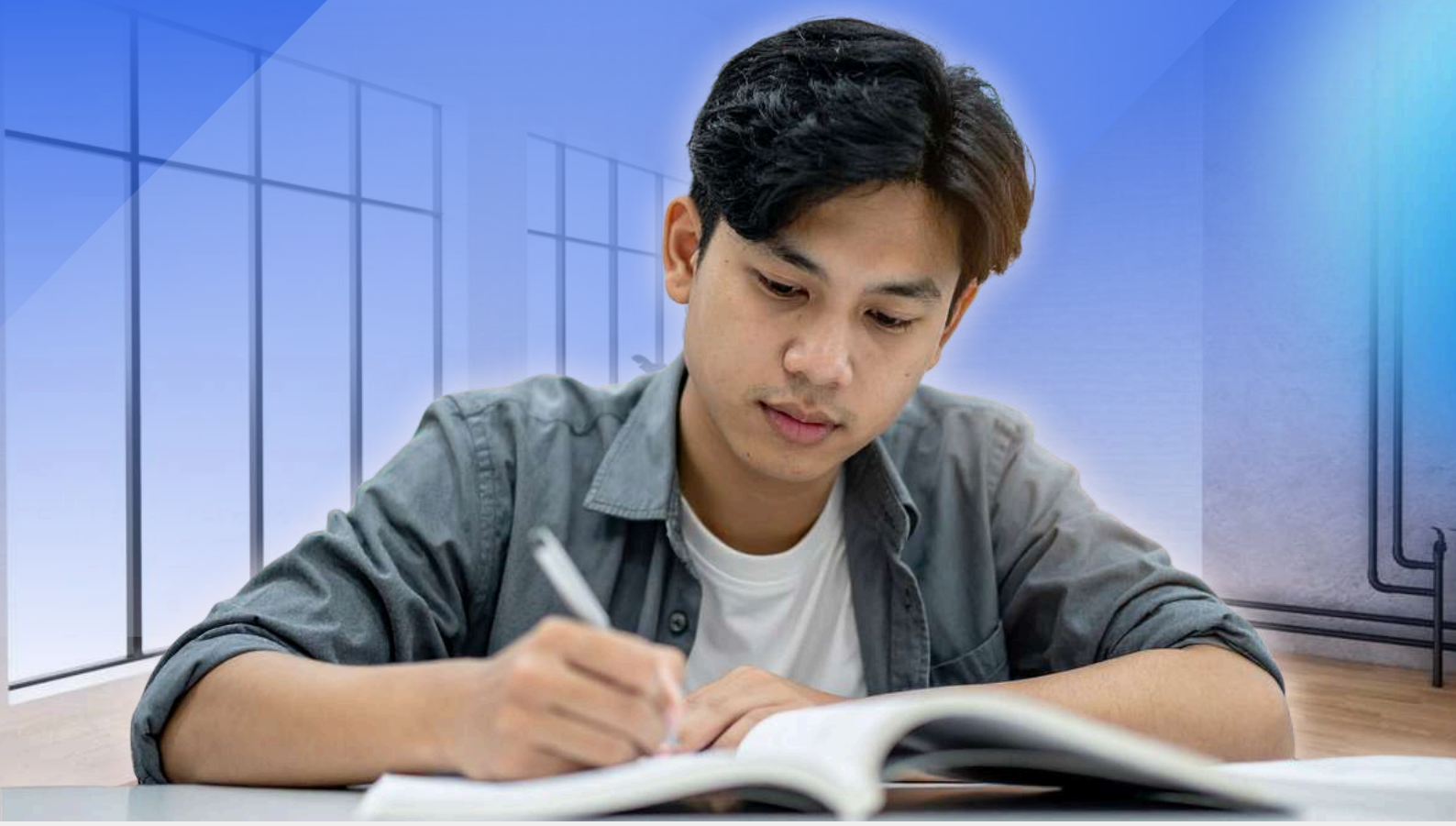
Lenterago

# TRY OUT #10

## SAT-Math

## 61010

44 QUESTIONS



www.lenterago.id
+6282170276076
office@lenterago.id

# MODUL 1 & 2

22 QUESTIONS

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

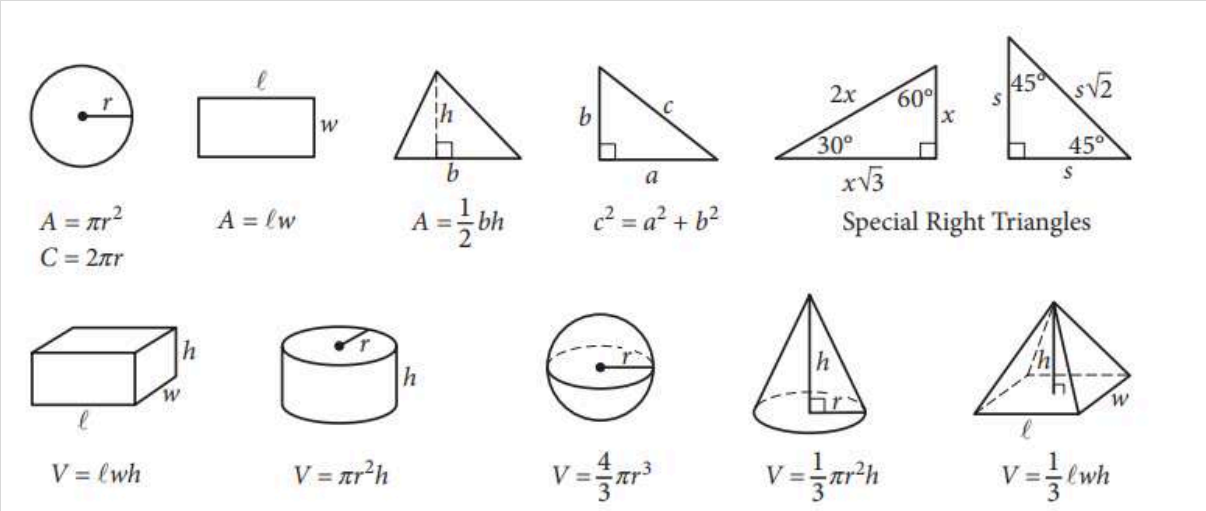
A. All variables and expressions represent real numbers.

B. Figures provided are drawn to scale.

C. All figures lie in a plane.

D. The domain of a given function *f* is the set of all real numbers *x* for which *f(x)* is a real number

REFERENCE




The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2p.

The sum of the measures in degrees of the angles of a triangle is 180.



082170276076

Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.

Module 1 & 2

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find more than one correct answer, write and circle only one answer.
- Your answer can be up to 5 characters for a positive answer and up to 6 characters (including the negative sign) for a negative answer, but no more.
- If your answer is a fraction that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a decimal that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a mixed number (such as 3 1/2), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include symbols such as a percent sign, comma, or dollar sign in your circled answer.



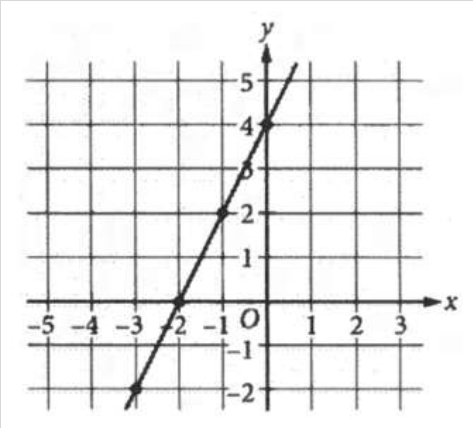
082170276076

Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.

1. If $3\left(\frac{x}{5}+\frac{1}{2}\right)+1=10$ what is the value of $2x+5$?
   a. 3
   b. 10
   c. 30
   d. 90

2. Harry is riding in a 100-mile-long bicycle race. The function $f$ defined by $f(x)=100-18x$ models the number of miles Harry has remaining x hours after he start the race. What is the best interpretation of $f(3)=46$ ?
   a. Harry rides about 46 mile in the first 3 hours of the race
   b. Harry will ride the last 46 mile of the race in the about 3 hours
   c. When Harry has ridden 46 miles, he has about 3 hours remaining
   d. Three hours after Harry start the race, he has about 46 miles remaining

3. The graph of $y=mx+b$, where $m$ and $b$ are constant is shown in the xy-plane.




   What is the value of $m+b$?
   a. 8
   b. 6
   c. 4
   d. 2

4.

| $x$ | $f(x)$ |
|---|---|
| $-4$ | 14 |
| $-2$ | 8 |
| 3 | $-7$ |
| 5 | $-13$ |

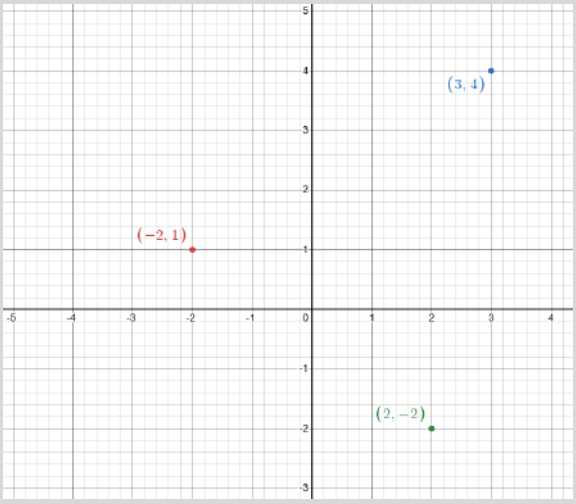
   For the linear function, $f$, the table show several value of x and their corresponding value of $f(x)$. What is the $y$ −intercept of the graph $y=f(x)$ in the $xy$ −plane?
   a. (0,-3)
   b. (0,-1)


082170276076

Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.

c. (0,2)
d. (0,4)

5.




Some coordinates are shown in the plane above. For which of the following inequality will contains all points as the solution?

a. $y \geq -x$
b. $y > -x - 2$
c. $y < x + 3$
d. $x < 3$

6.

$$y = \frac{1}{2}x + 5$$

One of the two equations in a linear system is given. The system has no solution. Which equation could be the second equation in this system?

a. $y = 2x + 5$
b. $y = 12x + 5$
c. $y = 12x - 5$
d. $y = -2x + 5$

7. In the xy-plane line $l$ has a slope of 2 . Line $k$ is perpendicular to line $l$ and contains point (4,2). Which following is an equation line $k$?

a. $y =- 2x - 6$
b. $y =- 2x + 10$
c. $y =- \frac{1}{2}x$
d. $y =- \frac{1}{2}x + 4$

8. The function of $f(x)$ is a linear function. The y-intercept of the graph of $y = f(x)$ in the xy-plane is (0,-12). What is the y-intercept of the graph $y = f(x) + 2$?

a. (0,-14)


082170276076

Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.

b. (0,-10)
c. (-2,-12)
d. (2,-12)

9.

$$2^{\frac{2}{n}}\left(\sqrt[n]{3}\right)$$

If n is a positive integer, which of the following is the equivalent to the expression above?

a. $3^{\frac{1}{n}}$
b. $18^{\frac{1}{n}}$
c. $\sqrt[n]{6}$
d. $\sqrt[n]{12}$

10. A psychologist conducting a memory experiment provided participants with a list of three-letter-sequences. Immediately after the experiment, the participants remember 100% of the sequences. The psychologist found that the percentage of sequences the participant remembered decreased by 20% for every 3-second interval that passed. Which function best models this situation, where P is the percentage of sequences the participants remembered, and t is the time, in second, that passed?

a. $P(t) = 100(0,20)^{3t}$
b. $P(t) = 100(0,20)^{t}$
c. $P(t) = 100(0,80)^{\frac{t}{3}}$
d. $P(t) = 100(0,80)^{t}$

11.

$$A = P(rt + 1)$$

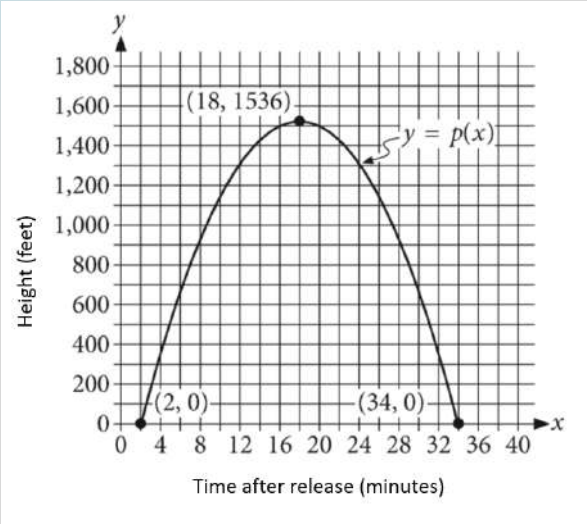
The equation shown gives A in the term of P, r and t where P and r are not equal to 0. Which equation gives t in the term of A, P and r?

a. $t = \frac{A}{P} - \frac{1}{r}$
b. $t = \frac{A}{Pr} - \frac{1}{Pr}$
c. $t = \frac{A}{Pr} - \frac{1}{r}$
d. $t = \frac{A}{r} - \frac{P}{r}$

12. A rocket is launched into the air. The function $h$ gives the total altitude $h(x)$, in feet, after the rocket has been in flight for x minutes. The graph of $y = h(x)$ represent the flight path of the rocket.


082170276076

Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.




What equation represent the relationship between the minutes and the height of the rockets?

a. $h(x) =- 6(x - 18)^2 + 34$

b. $h(x) =- 6(x - 18)^2 + 1536$

c. $h(x) = 6(x - 34)^2 + 18$

d. $h(x) = 6(x - 18)^2 + 1536$

13.

$$x^2(x + 3)(x - b) = 0$$

In the given equation, b is positive constant. The sum of the solution of the equation is 5. What is the value of b?

14.

$$x^2 - 4x - 9 = 0$$

The solutions to the given equation can be written in the form $\frac{m \pm \sqrt{k}}{2}$ where m and k are integers. What is the value of $\frac{k}{m}$?

15. The expression 1,4y represent the result of increasing the quantity y by p%. What is the value of p?

16. Each of whole numbers from 1 to 20 is written on one of 20 identical pieces of paper. The pieces of paper are then put into a hat and mixed together. If a piece of paper is selected at random from the hat, what is the probability that number on it is a multiple of 3?

a. $\frac{1}{20}$

b. $\frac{3}{20}$



082170276076

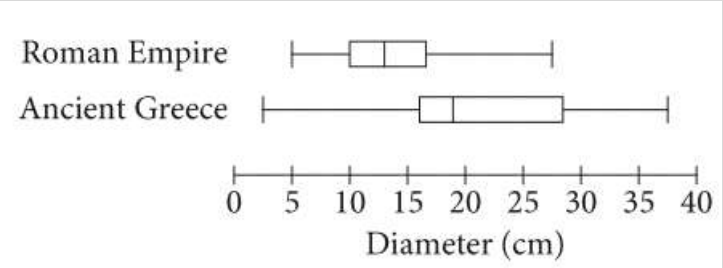
Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.

c. $\frac{6}{20}$

d. $\frac{1}{3}$

17. On January 1st, there are 2.500 boxes in a warehouse. On February 1st , there where 15% fewer boxes in the warehouse that there were on January 1st. On March 1st, there were 20% more boxes in the warehouse than there were on February 1st . What is the percentage of change from January 1st to March 1st?
    a. 2%
    b. 5%
    c. 10%
    d. 12,5%

18. The Metropolitan Museum of Art has plates on display from the Roman Empire and ancient Greece. The box plots shown summarize the distributions of the diameters, in centimeters, of all the museum's plates from each region.




What is the positive difference of range from both data?

19. The speed of a lightning bolt is approximately 320 million feet per second. What is the speed, in yards per minute? (1 yards = 3 feet)
    a. 160 million
    b. 960 million
    c. 6,400 million
    d. 19,200 million

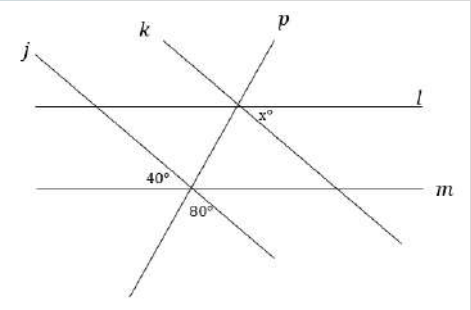
20. Two beach balls are each in the shape of sphere. The larger beach has a diameter 5x and the smaller has a diameter of 2x. What is the ratio of the volume of the larger beach ball to the volume of the smaller beach ball?
    a. 5 to 2
    b. 10 to 4
    c. 25 to 4
    d. 125 to 8

21.



082170276076

Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.




If the line j and k is parallel and l and m also parallel where p is a transversal. What is the value of x?

a. 20
b. 40
c. 60
d. 80

22. In a right triangle, the tangent of one of the two acute angles is $\frac{\sqrt{3}}{3}$. What is the tangent of the other acute angle?

a. $-\frac{\sqrt{3}}{3}$
b. $-\frac{3}{\sqrt{3}}$
c. $\frac{\sqrt{3}}{3}$
d. $\frac{3}{\sqrt{3}}$

23. Fiji has 1,500 yards of yarn. She wants to knit at least 2 scarves and at least 3 hats. Each scarf requires 300 yards of yarn, and each hat requires 120 yards of yarn. If s represents the number of scarves and h represent the number of hats, which of the following systems of inequality represent this situation?

a. $s + h \leq 1,500$
$s \geq 2$
$h \geq 3$
b. $2s + 3h \leq 1,500$
$s \geq 2$
$h \geq 3$
c. $2s + 3h \leq 1,500$
$s \geq 300$
$h \geq 120$
d. $300s + 120h \leq 1,500$
$s \geq 2$
$h \geq 3$


082170276076

Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.

24. If $10(x - 9) = 3(3x - 27) + 25$ what is the value of $(9 - x)$?
    a. -25
    b. -9
    c. 9
    d. 25

25. A school club is paid \$315 for recycling $c$ pounds of aluminum cans and p tons of paper. The equation $0,50c + 60p = 315$ represents this equation. What is the interpretation of $60p$ in this context?
    a. The total weight, in tons, of paper the club recycle
    b. The total amount, in dollars, the club is paid for each ton paper recycled
    c. The total weight, in tons, of all the aluminum cans and paper the club recycled
    d. The total amount, in dollars, the club paid for p tons of the paper recycled

26.
$$2a + b = 17$$
$$a + 2b = 19$$
The solution of the given system of equation is $(a, b)$. What is the value of $a + b$?
    a. 6
    b. 12
    c. 22
    d. 36

27. Ms. Anderson currently has 550 contacts on an online professional networking site. Her goal is to have at least 1,000 contacts. If she wants to meet this goal in 25 weeks, what is the minimum number of contacts per week, on average, she should add?
    a. 18
    b. 19
    c. 20
    d. 21

28. During an ice age, the average annual global temperature was at least 6 degrees Celsius lower than the modern average. If the average annual temperature of an ice age is y degrees Celsius and the modern average annual temperature is x degrees Celsius, which of the following must be true?
    a. $y = x - 6$
    b. $y \leq x + 6$
    c. $y \geq x - 6$
    d. $y \leq x - 6$

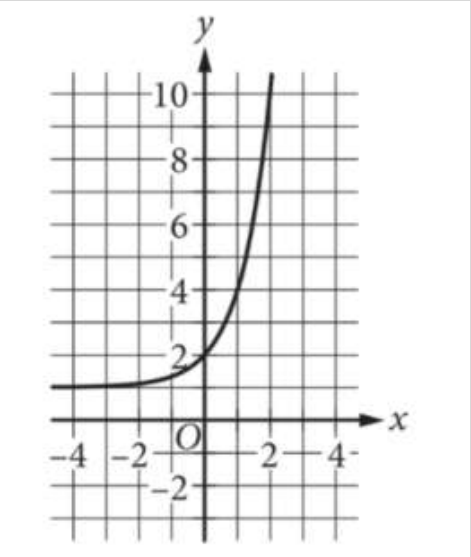
29. The system of equation above is graphed in the xy-plane. Which of the following is the y coordinate of an intersection of point (x,y) of the graph of the two equation?


082170276076

Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.

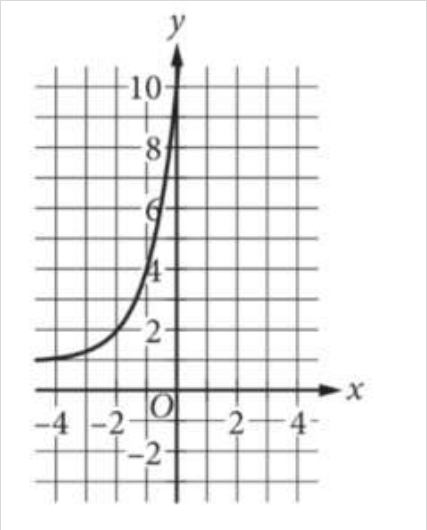
a. -4
b. -2
c. 2
d. 4

30.




The graph of $y = f(x) + 2$ is shown as below. What is the graph of $f(x)$?

a.

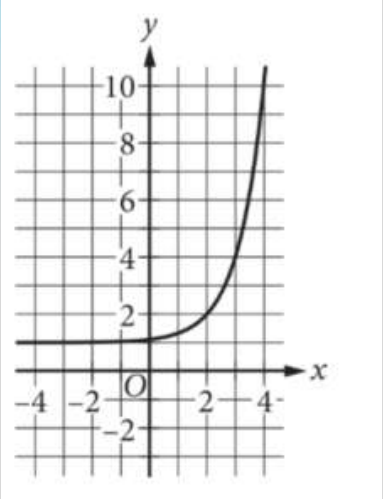




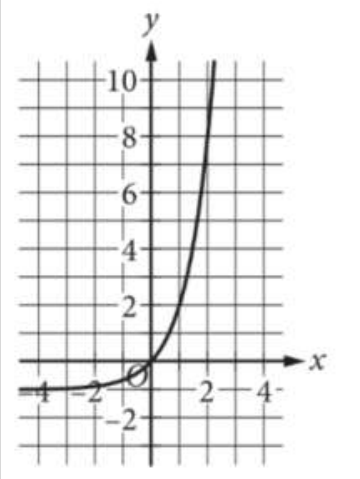
082170276076

Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.

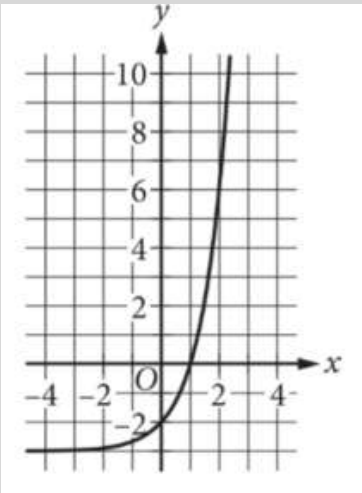
b.




c. .




d. .






082170276076

Dokumen ini bersifat **RAHASIA DAN MILIK INTELEKTUAL LENTERAGO**.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.

31. When an All You Can Eat restaurant charges $12.00 per meal, the number of meals it sells per day is 400 pax. For each $0,50 increased to the price per meal, the number of meals sold per day decreased by 10. What is the price per meal that result the greatest sales, in dollars, from meal each day?
    a. $ 16.00
    b. $20.00
    c. $24.00
    d. $28.00

32.

$$x^2 + bx + 16 = 0$$

In the quadratic equation shown, b is a constant. For what value of b does the equation have only one solution?
    a. -4
    b. -8
    c. -4 and 4
    d. -8 and 8

33.

$$L_1 = 10(0.89)^d$$
$$L_2 = 9.75 - 0.83d$$

An exponential equation and a linear equation are given. Each equation estimate the luminosity, in units of billion of solar luminosities, for a Type Ia supernova d days after its peak luminosity, for $d \leq 6$. The luminosity, in billions of solar luminosities, estimated by the linear equation 3 days after its peak luminosity is how much greater than the luminosity estimates by exponential equation?
    a. 0.21
    b. 2.2
    c. 21
    d. 210

34. If $ax - 3$ is a factor of $6x^3 + 27x^2 - 54x$, where a is a positive constant, what is the value of a?
    a. 1
    b. 2
    c. 3
    d. 6

35.

$$t = \frac{5}{x-1} + \frac{8}{2x-2}$$


082170276076

Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.

Which of the following express x in the term of t?

a. $\frac{9}{t-1}$

b. $\frac{9+t}{t}$

c. $t - \frac{18}{t}$

d. $\frac{9-t}{t-1}$

36. Which of the following is equivalent to the expression $x^4 - 18x^2 + 81$ ?

I. $(x + 3)^2(x - 3)^2$

II. $\left(x^2 - 9\right)^2$

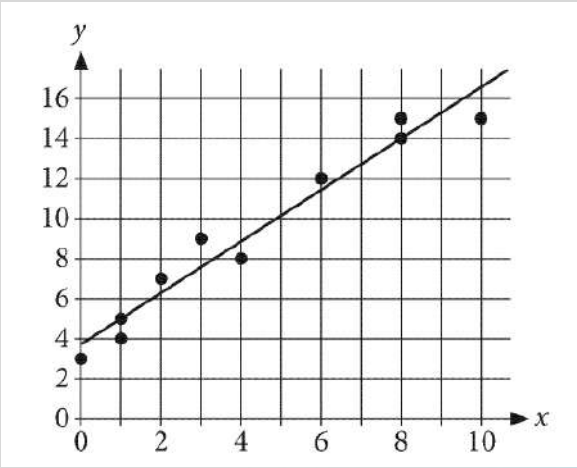
a. I only
b. II only
c. I and II
d. Neither I or II

37.

$$0.10x + 0.20y = 0.18(x + y)$$

The given equation represent a volume $x$, in gallons of a 10% saltwater solution that will be mixed with a volume $y$, in gallons, of a 20% saltwater solution to produce 18% saltwater solution. What volume, in gallons, of the 10% saltwater will be needed if 200 gallons of 20% saltwater is used?

38. The scatterplot show the relationship between two variable, x and y. A line of best fit is also shown.




What is the equation of the line of best fit?

a. $y = 3.7 + 1.3x$

b. $y = 1.3 + 3.7x$

c. $y = 3.7 - 1.3x$



082170276076

Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.

d. $y = 1.3 - 3.7x$

39.

0,2,3,4,5,5,5,6,6,7

The given list shows a baseball team's score for each of its first 10 games. In the eleventh game, the team has score of 18. Which of the following best describe the mean and the median of the team's for the first 11 games compared to the first 10 games?

a. Mean increased and the median remain unchanged
b. The median increased and the mean remained unchanged
c. Both the mean and the median remain unchanged
d. Both the mean and the median increased

40.

| Age group (in year) | Agree | Disagree | Total |
|---|---|---|---|
| 18-29 | 113 | 109 | 222 |
| 30-44 | 126 | 136 | 262 |
| 45-64 | 145 | 201 | 346 |
| 65 and up | 68 | 102 | 170 |
| Total | 452 | 548 | 1000 |

The table show the result of a poll of 1,000 people when asked to agree or disagree with the statement "I play too much on my phone daily" . If the respondent who was selected at random disagree with the statement, which of the following is the closest to the probability that the responded selected is at least 45 years old?

a. 0.37
b. 0.45
c. 0.49
d. 0.55

41. A contractor purchased two slabs of granite, both in shape of a right rectangular prism. The table below show some information about two slabs.

| | Length | Width | Thickness | Mass |
|---|---|---|---|---|
| Slab 1 | 100 centimeters | 20 centimeters | 8 centimeters | 44,000 grams |
| Slab 2 | 125 centimeters | | 8 centimeters | |

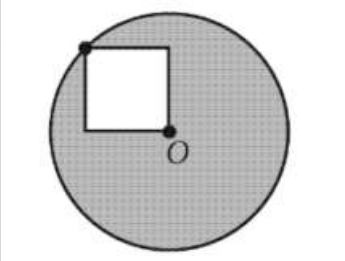
What is the density, in grams per cubic centimeter, of Slab 1?



082170276076

Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.

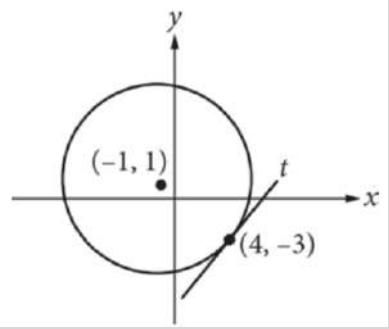
42.




In the figure shown , point O is the center of the circle. One vertex of the square lies on the circle and the opposite vertex is point O. If the area of the shaded region is $36\pi - 18$. What is the area of the square?

a. $3\sqrt{2}$
b. $12\sqrt{2}$
c. 18
d. 24

43.



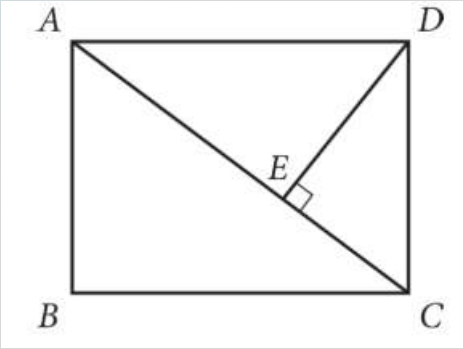

The circle has center (-1,1). Line t is a tangent to this circle at point (4,-3). Which of the following the equation of the circle?

a. $(x - 1)^2 + (y + 1)^2 = 41$
b. $(x + 1)^2 + (y - 1)^2 = 41$
c. $(x - 4)^2 + (y + 3)^2 = 41$
d. $(x + 4)^2 + (y - 1)^2 = 41$

44.


082170276076

Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.




In rectangle ABCD shown, $AB = 6$, $CD = 6$ and $AC = 10$. What is the length of $CE$?

a. 3.6
b. 4.2
c. 5
d. 6


082170276076

Dokumen ini bersifat RAHASIA DAN MILIK INTELEKTUAL LENTERAGO.
Segala bentuk reproduksi atau distribusi tanpa izin tertulis dapat ditindak sesuai hukum yang berlaku.